

THE KIANI–TAVAKOLIPOUR DICHOTOMY AT $m = 5$ FOR SYMMETRIC TROPICAL MATRICES, IN ADDITIVE FORM

ABSTRACT. For an $n \times n$ matrix A over $\mathbb{R}_{\max} = \mathbb{R} \cup \{-\infty\}$ let $\delta_k(A)$ be the largest tropical permanent of a principal $k \times k$ submatrix of A , with $\delta_0 = 0$; these are the coefficients of the tropical characteristic polynomial. Kiani and Tavakolipour conjecture that for symmetric A and every m with $3 \leq m \leq n$ at least one of $\delta_m - \delta_{m-1} \leq \delta_{m-1} - \delta_{m-2}$ and $\delta_m - \delta_{m-2} \leq 2(\delta_{m-2} - \delta_{m-3})$ holds; they prove the cases $m = 3$ and $m = 4$ and leave $m \geq 5$ open. We prove that for every $n \geq 5$ and every symmetric A over $\mathbb{R}_{\max}$, at least one of the additive inequalities $\delta_5 + \delta_3 \leq 2\delta_4$ and $\delta_5 + 2\delta_2 \leq 3\delta_3$ holds. Whenever the δ_k involved are finite these are equivalent to the two conjectured inequalities at $m = 5$, so the conjecture is settled at $m = 5$ in every case in which its differences are defined; when some $\delta_k = -\infty$ the additive inequalities are our chosen subtraction-free extension of the source statement rather than that statement itself. The failure of both additive inequalities is first rewritten as a system of strict homogeneous linear inequalities in the entries of A ; a nonnegative rational combination of those forms that vanishes identically then refutes the failure. A failure is witnessed by three optimal selections of sizes 5, 3 and 2, whose union meets at most 10 indices, and entrywise nonnegativity shows that a certificate may be sought inside that index set, which handles all n at once. What remains is a finite check: $7 \times 600 \times 90 = 378,000$ triples, falling into 53,900 classes with identical constraint systems, each carrying such a combination; 38,524 of the classes are killed by a two-term identity between multisets of matrix positions, the rest by longer identities or a general rational combination. The cases $m \geq 6$ remain open.

1. THE CONJECTURE

Let $\mathbb{R}_{\max} = \mathbb{R} \cup \{-\infty\}$ with $a \oplus b = \max(a, b)$ and $a \odot b = a + b$. For a $k \times k$ matrix B over $\mathbb{R}_{\max}$ the tropical permanent is $\text{per}(B) = \max_{\pi \in S_k} \sum_{i=1}^k b_{i\pi(i)}$. For an $n \times n$ matrix A over $\mathbb{R}_{\max}$ and $0 \leq k \leq n$ set

$$(1) \quad \delta_k(A) = \max_{S \subseteq [n], |S|=k} \text{per}(A[S]), \quad \delta_0(A) = 0,$$

where $A[S]$ is the principal submatrix on the rows and columns indexed by S . Up to sign these are the coefficients of the tropical characteristic polynomial of A ; (1) is the description used throughout [2, Thm. 2.6 and eq. (2.3)]. We write δ_k for $\delta_k(A)$.

2020 *Mathematics Subject Classification.* 15A80, 14T90, 90C05.

Key words and phrases. tropical characteristic polynomial, max-plus algebra, symmetric matrix, principal submatrix, tropical permanent, concavity, Gordan's theorem.

The conjecture below is Conjecture 6.1 of [2], stated there for “the sequence (δ_m) associated with *symmetric* tropical matrices”. It reads, verbatim:

Conjecture 1 (Kiani–Tavakolipour [2, Conj. 6.1]). *For $3 \leq m \leq n$, at least one of the following inequalities holds:*

$$(2) \quad \delta_m - \delta_{m-1} \leq \delta_{m-1} - \delta_{m-2}, \quad \text{or}$$

$$(3) \quad \delta_m - \delta_{m-2} \leq 2(\delta_{m-2} - \delta_{m-3}).$$

The additive reading. The semiring is $\mathbb{R}_{\max}$, so a matrix may have $-\infty$ entries and a δ_k may equal $-\infty$; in that case the differences in (2) and (3) have no literal meaning, and we make no claim about what Conjecture 1 asserts there. We work instead with the following additive forms, obtained by moving the negative terms to the other side. These are the statements we prove:

$$(I) \quad \delta_m + \delta_{m-2} \leq 2\delta_{m-1},$$

$$(II) \quad \delta_m + 2\delta_{m-3} \leq 3\delta_{m-2}.$$

Whenever all the δ_k involved are finite, (I) is equivalent to (2) and (II) to (3); outside that case (I)–(II) are our chosen extension of the source’s subtraction-based statement, not a reading we attribute to [2].

Conjecture 1 is proved in [2] for $m = 3$ (their Thm. 4.1) and $m = 4$ (their Thm. 4.5); $m = 5$ is the first case the source leaves open.

Theorem 2. *Let $n \geq 5$ and let A be a symmetric $n \times n$ matrix over $\mathbb{R}_{\max}$. Then*

$$\delta_5 + \delta_3 \leq 2\delta_4 \quad \text{or} \quad \delta_5 + 2\delta_2 \leq 3\delta_3.$$

That is, at least one of (I), (II) holds at $m = 5$, for every n and with no finiteness assumption on A . Whenever $\delta_5, \delta_4, \delta_3, \delta_2$ are finite this is exactly the assertion of Conjecture 1 at $m = 5$, so the conjecture is settled at $m = 5$ in every case in which its differences are defined; in the remaining cases the theorem asserts the additive reading (I)–(II) and not the source statement.

2. COVERS

A k -cover of $[n]$ is a pair $P = (S, \pi)$ with $S \subseteq [n]$, $|S| = k$, and π a permutation of S ; write $\text{supp } P = S$ and $w(P) = \sum_{i \in S} a_{i\pi(i)} \in \mathbb{R}_{\max}$. Let $\mathcal{C}_k$ be the set of k -covers. Reading (1) through the permanent gives

$$(4) \quad \delta_k = \max_{P \in \mathcal{C}_k} w(P),$$

the point being that a principal submatrix forces the row set and the column set to coincide. The unique 0-cover has weight 0, consistent with $\delta_0 = 0$.

Since A is symmetric, $w(P)$ depends on P only through the multiset

$$E(P) = \{\{i, \pi(i)\} : i \in \text{supp } P\}$$

of *positions* $\{i, j\}$ (with $i = j$ allowed), which we regard as a vector of nonnegative integers indexed by positions; then $w(P) = \sum_p E(P)_p a_p$. A loop $i \mapsto i$ contributes $\{i, i\}$ once; a transposition (ij) contributes $\{i, j\}$ twice, and indeed weight $2a_{ij}$. Write $\text{pos } P$ for the *set* of positions occurring in $E(P)$. A k -cycle and its inverse have the same E , so for symmetric A the relevant object is $E(P)$, not P .

3. THE FAILURE OF BOTH INEQUALITIES, AS A LINEAR SYSTEM

Throughout this section $m = 5$; nothing in the argument is special to 5.

Lemma 3. *Both (I) and (II) fail at a symmetric A if and only if there are $P \in \mathcal{C}_5$, $Q \in \mathcal{C}_3$, $S \in \mathcal{C}_2$ with*

$$\begin{aligned} (5) \quad & w(P) + w(Q) > 2w(R) && \text{for every } R \in \mathcal{C}_4, \\ (6) \quad & w(P) + 2w(S) > 3w(Q') && \text{for every } Q' \in \mathcal{C}_3. \end{aligned}$$

Moreover $w(P), w(Q), w(S)$ are then finite.

Proof. Suppose both fail, i.e. $\delta_5 + \delta_3 > 2\delta_4$ and $\delta_5 + 2\delta_2 > 3\delta_3$. In $\mathbb{R}_{\max}$ a left-hand side equal to $-\infty$ exceeds nothing, so $\delta_5, \delta_3, \delta_2$ are finite. Take P, Q, S attaining the maxima in (4) for $k = 5, 3, 2$; their weights are $\delta_5, \delta_3, \delta_2$, finite. For $R \in \mathcal{C}_4$, $2w(R) \leq 2\delta_4 < \delta_5 + \delta_3 = w(P) + w(Q)$, which is (5); (6) is identical with δ_3, δ_2 in place of δ_4, δ_3 .

Conversely, given such P, Q, S , the right-hand sides of (5) and (6) already range over everything the maxima δ_4, δ_3 range over, so $2\delta_4 < w(P) + w(Q) \leq \delta_5 + \delta_3$ and $3\delta_3 < w(P) + 2w(S) \leq \delta_5 + 2\delta_2$, using $\delta_5 \geq w(P)$, $\delta_3 \geq w(Q)$, $\delta_2 \geq w(S)$ and monotonicity of $\oplus$ -addition. Finiteness of $w(P), w(Q), w(S)$ is forced by (5) and (6) themselves. $\square$

Fix such a triple and set

$$\mathcal{A} = \text{pos } P \cup \text{pos } Q \cup \text{pos } S, \quad D = |\mathcal{A}| \leq 5 + 3 + 2 = 10.$$

Because $w(P), w(Q), w(S)$ are finite, *every entry a_p with $p \in \mathcal{A}$ is finite*. Put

$$\text{Pool}_4 = \{R \in \mathcal{C}_4 : \text{pos } R \subseteq \mathcal{A}\}, \quad \text{Pool}_3 = \{Q' \in \mathcal{C}_3 : \text{pos } Q' \subseteq \mathcal{A}\},$$

where multiplicities are unconstrained: a transposition on $\{i, j\}$ uses the single position $\{i, j\}$ twice and belongs to the pool as soon as $\{i, j\} \in \mathcal{A}$. For $R \in \text{Pool}_4$ and $Q' \in \text{Pool}_3$ define integer vectors in $\mathbb{Z}^{\mathcal{A}}$,

$$(7) \quad f_R = E(P) + E(Q) - 2E(R), \quad g_{Q'} = E(P) + 2E(S) - 3E(Q').$$

Evaluated at the vector $x = (a_p)_{p \in \mathcal{A}} \in \mathbb{R}^D$ of *finite* entries, $f_R(x) = w(P) + w(Q) - 2w(R)$ and $g_{Q'}(x) = w(P) + 2w(S) - 3w(Q')$; so (5) and (6), restricted to the pool, say exactly that x makes every form in (7) strictly positive.

Lemma 4 (certificates). *Let $E(P), E(Q), E(S)$ be given, with $\mathcal{A}$ and the pools as above. Suppose there are nonnegative rationals λ_R ($R \in \text{Pool}_4$) and $\mu_{Q'}$ ($Q' \in \text{Pool}_3$), not all zero, with*

$$(8) \quad \sum_R \lambda_R f_R + \sum_{Q'} \mu_{Q'} g_{Q'} = 0 \quad \text{in } \mathbb{Q}^{\mathcal{A}}.$$

Then no symmetric A , over any index set containing $\mathcal{A}$, satisfies (5) and (6) for this triple.

Proof. If it did, all entries indexed by $\mathcal{A}$ would be finite by Lemma 3 and each form would take a strictly positive real value at x ; evaluating (8) at x gives $0 = \sum \lambda_R f_R(x) + \sum \mu_{Q'} g_{Q'}(x) > 0$, since the coefficients are nonnegative and not all zero. Contradiction. $\square$

Normalising $\sum \lambda_R + \sum \mu_{Q'} = 1$ and writing $L = \sum \lambda_R$, $M = \sum \mu_{Q'}$, (8) is the identity of position multisets

$$(9) \quad E(P) + L E(Q) + 2M E(S) = 2 \sum_R \lambda_R E(R) + 3 \sum_{Q'} \mu_{Q'} E(Q'),$$

with $L + M = 1$, $L, M \geq 0$. Suppose A is symmetric and P, Q, S attain the maxima in (4) for $k = 5, 3, 2$, so that $w(P) = \delta_5$, $w(Q) = \delta_3$ and $w(S) = \delta_2$, all finite, as for the triple supplied by Lemma 3. Evaluating (9) at A and using $\delta_4 \geq w(R)$ for $R \in \text{Pool}_4$, $\delta_3 \geq w(Q')$ for $Q' \in \text{Pool}_3$, together with $\sum_R \lambda_R = L$, $\sum_{Q'} \mu_{Q'} = M$ and $L + M = 1$, then gives $L(\delta_5 + \delta_3 - 2\delta_4) + M(\delta_5 + 2\delta_2 - 3\delta_3) \leq 0$, so one of (I), (II) holds. Checking one certificate requires only that the multipliers be nonnegative and that a rational linear identity hold; but the multipliers are printed below only for the sample certificates, so for the rest of the census that check is available only by running the program. No linear-programming duality is needed: only the trivial direction of Gordan's theorem [1] is used, namely that (8) obstructs a strictly positive solution; the converse is not used.

4. ALL n AT ONCE: THE WINDOW $[3m - 5]$

Lemma 5. *Suppose both (I) and (II) fail at some symmetric A over $\mathbb{R}_{\max}$ of some size $n \geq 5$. Then there are $P_0 \in \mathcal{C}_5([5])$, $Q_0 \in \mathcal{C}_3([10])$, $S_0 \in \mathcal{C}_2([10])$, with P_0 a representative of its cycle type, such that the triple $(E(P_0), E(Q_0), E(S_0))$ admits no certificate (8).*

Proof. Take P, Q, S from Lemma 3. Their supports have union of size at most $5 + 3 + 2 = 10$, and $|\text{supp } P| = 5$. Choose a bijection of that union with a subset of $[10]$ carrying $\text{supp } P$ onto $[5]$, then compose with a permutation of $[5]$, extended by the identity on $\{6, \dots, 10\}$, carrying P to the chosen representative of its cycle type. Relabelling the index set carries $\mathcal{A}$, Pool_4 and Pool_3 bijectively onto the relabelled ones and permutes the coordinates of every form in (7), so a certificate for the image triple would be a certificate for the original, contradicting Lemma 4 together with (5)–(6). The images Q_0, S_0 are covers of $[10]$ of sizes 3 and 2. $\square$

The window comes from the union of the supports of P , Q and S : $|\mathcal{A}| \leq m + (m-2) + (m-3) = 3m-5$, which is 10 at $m = 5$. Entrywise nonnegativity is what makes it enough to search there. In the identity (9) every vector has nonnegative entries, so at a position outside $\mathcal{A}$ the left side is 0 and therefore every R with $\lambda_R > 0$ and every Q' with $\mu_{Q'} > 0$ vanishes there; a certificate never leaves the window, so restricting the pools to it loses nothing.

5. THE FINITE CHECK

By Lemma 5, Theorem 2 follows once a certificate is exhibited for every triple $(E(P_0), E(Q_0), E(S_0))$ in the following finite list. The list is completely specified by the three counts below, and a reader can regenerate it without any code.

P_0 : *seven triples' worth of cycle type*. Two 5-covers of a 5-set have relabelling-equivalent position multisets precisely when they have the same cycle type, since a cycle and its inverse have the same E and relabelling preserves cycle type. So P_0 runs over the **seven** partitions of 5,

$$1^5, \quad 2 \cdot 1^3, \quad 2^2 \cdot 1, \quad 3 \cdot 1^2, \quad 3 \cdot 2, \quad 4 \cdot 1, \quad 5.$$

Q_0 : *the 600 distinct E of a 3-cover of [10]*. Three loops, $\binom{10}{3} = 120$; a transposition and a loop, $\binom{10}{2} \cdot 8 = 360$; a 3-cycle, one position multiset per 3-subset because a 3-cycle and its inverse agree, $\binom{10}{3} = 120$. Total $120 + 360 + 120 = 600$.

S_0 : *the 90 distinct E of a 2-cover of [10]*. Two loops, $\binom{10}{2} = 45$; a transposition, 45. Total 90.

The list therefore has

$$7 \times 600 \times 90 = 378,000$$

entries. Every one of them carries a certificate (8); none is live. This is the computational content of the paper and it is what `verify.py` in the accompanying folder re-derives from scratch. Under the order-preserving relabelling of the index set the 378,000 triples fall into 53,900 classes with identical constraint systems, and the program solves one system per class.

The certificates are small and mostly of two shapes. Writing ij for the position $\{i, j\}$:

- 38,524 of the 53,900 classes are killed by a *two-term identity in branch (I)*: two pool 4-covers R_1, R_2 with

$$E(P) + E(Q) = E(R_1) + E(R_2),$$

i.e. (8) with $\lambda_{R_1} = \lambda_{R_2} = \frac{1}{2}$, $L = 1$, $M = 0$.

- 1,016 classes are killed by a *three-term identity in branch (II)*: pool 3-covers with $E(P) + 2E(S) = E(Q'_1) + E(Q'_2) + E(Q'_3)$, i.e. $\mu_{Q'_i} = \frac{1}{3}$, $L = 0$, $M = 1$.

- the remaining 14,360 classes need a general nonnegative rational combination. Over all 53,900 classes the largest denominator occurring in any multiplier is 48, so every certificate is a small rational vector.

Two certificates in full, hand-checkable by counting symbols. In each, P is the all-loops 5-cover of $\{0, 1, 2, 3, 4\}$.

$$\begin{aligned} E(P) + E(Q) &= \{00, 11, 22, 33, 44\} + \{00, 11, 22\} \\ &= \{00, 00, 11, 11, 22, 22, 33, 44\} \\ &= \{00, 11, 22, 33\} + \{00, 11, 22, 44\} = E(R_1) + E(R_2), \end{aligned}$$

with Q the three loops on $\{0, 1, 2\}$ and R_1, R_2 the all-loops 4-covers of $\{0, 1, 2, 3\}$ and $\{0, 1, 2, 4\}$; and, with Q the transposition $(0\ 1)$ together with the loop at 2,

$$\begin{aligned} E(P) + E(Q) &= \{00, 11, 22, 33, 44\} + \{01, 01, 22\} \\ &= \{01, 01, 22, 33\} + \{00, 11, 22, 44\} = E(R_1) + E(R_2), \end{aligned}$$

where now R_1 is the transposition $(0\ 1)$ with loops at 2, 3. In both cases $L = 1$, $M = 0$ and the two multipliers are $\frac{1}{2}$, so (9) gives $\delta_5 + \delta_3 \leq 2\delta_4$ at A ; that is, branch (I) holds.

Proof of Theorem 2. If the conclusion failed for some symmetric A and some $n \geq 5$, then by Lemma 5 some triple in the list above would admit no certificate (8). Every triple in the list does admit one. Hence no such A exists. $\square$

6. WHAT IS NOT SETTLED

$m \geq 6$. Theorem 2 settles the single value $m = 5$. Conjecture 1 ranges over $3 \leq m \leq n$ and remains open for every $m \geq 6$. The reduction above is generic in m and gives the window $[3m - 5] = [13]$ at $m = 6$, with a raw count of $11 \times |\mathcal{C}_4([13])| \times |\mathcal{C}_3([13])| = 11 \times 12,155 \times 1,430 \approx 1.9 \times 10^8$ triples, which we have not attempted. Nothing here suggests the answer at $m \geq 6$ either way.

Nonsymmetric matrices. Symmetry is essential and [2] says so; Theorem 2 makes no claim about nonsymmetric A .

The multiplier vectors are not shipped. Only the two certificates above appear in print. The multipliers for the rest of the 53,900 classes exist only inside a run of `verify.py`, which regenerates and re-verifies them but writes no file listing them, so the census is reproducible without being auditable from data in this folder.

Only one direction of Gordan's theorem is used. We never claim that a triple *without* a certificate yields a genuine counterexample.

7. RELATION TO EARLIER WORK

Lewis [3] studies the same object under the name *best principal submatrix problem*, settling the symmetric case with entries in $\{0, -\infty\}$ and the symmetric case whose digraph has no odd cycle; neither states an inequality of the shape (2)–(3). Rosenmann [4] proves concavity in k for the bipartite k -cardinality assignment problem, where the k rows and the k columns are chosen independently, whereas (1) forces the two sets to coincide.

8. REPRODUCTION AND VERIFICATION

Sections 2–5 are self-contained: the census is specified by the seven cycle types, $\mathcal{C}_3([10])$ and $\mathcal{C}_2([10])$, and a reader can regenerate it in a few lines and search for the certificates directly. The accompanying `verify.py` does exactly that, in Python 3 with the standard library only and in exact integer and `Fraction` arithmetic. It re-derives the counts 7, 600, 90 and 378,000; checks the printed certificates as exact multiset identities; and re-runs the $m = 5$ census, re-substituting every one of the 53,900 multiplier vectors it finds into its own forms. It closes with VERDICT: ALL 47 CHECKS PASS; `verify.output.txt` in the same folder records one such run.

The program prints its own limits: it does not touch $m \geq 6$, it does not persist the multiplier vectors it verifies, and it makes no literature claim.

REFERENCES

- [1] P. Gordan, *Über die Auflösung linearer Gleichungen mit reellen Coefficienten*, Math. Ann. **6** (1873), 23–28. doi:10.1007/BF01442864.
- [2] D. Kiani and H. Tavakolipour, *Properties of the tropical characteristic polynomial of symmetric matrices*, arXiv:2607.10922v1, 12 July 2026. arXiv:2607.10922. Statement numbers are those of that e-print, the only text we could access.
- [3] S. C. Lewis, *On the best principal submatrix problem*, Ph.D. thesis, University of Birmingham, 2007. theses.bham.ac.uk/id/eprint/26.
- [4] A. Rosenmann, *Computing the sequence of k -cardinality assignments*, J. Comb. Optim. **44** (2022), 1265–1283. doi:10.1007/s10878-022-00840-7; arXiv:2104.04037.